

Connect and Protect: Networks and Network Security

Module 1
Network architecture

Module 2
Network operations

Module 3
Secure against network intrusions

Introduction to network intrusion tactics

- Welcome to module 3
 - Video • 25 sec
- The case for securing networks
 - Video • 1 min
- How intrusions compromise your system
 - Reading • 4 min
- Matt: A professional on dealing with attacks
 - Video • 3 min

Secure networks against Denial of Service (DoS) attacks

- Denial of Service (DoS) attacks
 - Video • 4 min
- Read tcpdump logs
 - Reading • 4 min
- Real life DoS attack
 - Reading • 4 min
- Activity: Analyze network layer communication
 - Practice Assignment • Grade: 100%
- Activity Exemplar: Analyze network layer communication
 - Reading • 4 min
- Test your knowledge: Secure networks against Denial of Service (DoS) attacks
 - Practice Assignment • Grade: 93.75%

Network attack tactics and defense

- Malicious packet sniffing
 - Video • 4 min
- IP Spoofing
 - Video • 2 min
- Overview of interception tactics
 - Reading • 8 min
- Identify: Network attacks
 - Ungraded Plugin • 10 min
- Activity: Analyze network attacks
 - Practice Assignment • Grade: 100%
- Activity Exemplar: Analyze network attacks
 - Reading • 4 min
- Test your knowledge: Network interception attack tactics
 - Practice Assignment • Grade: 75%

Review: Secure against network intrusion

- Wrap up
 - Video • 44 sec
- Glossary terms from module 3
 - Reading • 4 min
- Module challenge 3
 - Graded Assignment • 50 min

Module 4
Security hardening

Glossary terms from module 3

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Terms and definitions from Course 3, Module 3

Active packet sniffing: A type of attack where data packets are manipulated in transit

Botnet: A collection of computers infected by malware that are under the control of a single threat actor, known as the "bot-herder"

Denial of service (DoS) attack: An attack that targets a network or server and floods it with network traffic

Distributed denial of service (DDoS) attack: A type of denial of service attack that uses multiple devices or servers located in different locations to flood the target network with unwanted traffic

Internet Control Message Protocol (ICMP): An internet protocol used by devices to tell each other about data transmission errors across the network

Internet Control Message Protocol (ICMP) flood: A type of DoS attack performed by an attacker repeatedly sending ICMP request packets to a network server

IP spoofing: A network attack performed when an attacker changes the source IP of a data packet to impersonate an authorized system and gain access to a network

On-path attack: An attack where a malicious actor places themselves in the middle of an authorized connection and intercepts or alters the data in transit

Packet sniffing: The practice of capturing and inspecting data packets across a network

Passive packet sniffing: A type of attack where a malicious actor connects to a network hub and looks at all traffic on the network

Ping of death: A type of DoS attack caused when a hacker pings a system by sending it an oversized ICMP packet that is bigger than 64KB

Replay attack: A network attack performed when a malicious actor intercepts a data packet in transit and delays it or repeats it at another time

Snarf attack: A network attack performed when an attacker sniffs an authorized user's IP address and floods it with ICMP packets

Synchronize (SYN) flood attack: A type of DoS attack that simulates a TCP/IP connection and floods a server with SYN packets

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Explain this topic in simple terms

Give me a summary

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https://www.coursera.org/learn/networks-and-network-security/supplement/xiHgn/glossary-terms-from-module-3

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